Project Summary: Cart Abandonment Analysis

Business Question: "Where in the checkout flow are customers dropping off most, and what should the product team fix first?"

> **Dataset Scale:** 67.4M Cleaned Clickstream Events | 1.74M Cart Sessions | 13.77M User Sessions
> **Core Financial Impact:** $381.90M Total Potential Lost GMV | 62.81% Cart Abandonment Rate
> **Analytics Tech Stack:** Python (Pandas) | MySQL (Aggregations, CTEs) | Power BI (DAX, UI/UX)

1. Project Context & Objectives

Cart abandonment is one of the costliest points of revenue leakage in modern eCommerce platforms. While millions of prospective buyers browse catalogs and add items to digital carts, only a fraction complete the final transaction. This project investigates the complete November 2019 eCommerce user interaction journey to diagnose precisely where prospective customers abandon checkout and quantify the gross merchandise value (Lost GMV) at stake.

2. Diagnostic Findings: Where Customers Are Dropping Off Most

Through data modeling in MySQL and interactive visualization in Power BI, the root leakage points were isolated across four critical analytical dimensions:

Cart-to-Purchase Bottleneck (Primary Operational Leakage)

While the top-of-funnel view-to-cart drop-off is 87.3% (typical for browsing behavior), the critical commercial breakdown occurs between Cart and Purchase. Exactly 62.81% of cart sessions (1.09M sessions) drop off before checkout completion. Only 648.36K purchases were finalized out of 1.74M cart sessions, leading directly to a massive $381.90M in Potential Lost GMV.

Category & Brand Concentration (Electronics & Smartphones Dominance)

Revenue loss is intensely concentrated in high-value personal tech. The Electronics category alone accounts for $258.37M (67.6%) of all lost revenue. Drilling deeper, Smartphones represent $218.93M of that leakage. At the brand level, Apple ($160.55M) and Samsung ($64.24M) together account for over 58% of platform-wide lost GMV, proving that abandonment is driven by high-intent tech shoppers.

Price Sensitivity Curve ($150 to $1,000 High-Value Gaps)

Although items priced Under $50 exhibit a high drop-off rate (62.51%), their total monetary impact is negligible ($8M). In stark contrast, 67% of all lost GMV is concentrated in two mid-to-premium brackets: $150–$500 ($132M Lost GMV, 57.01% drop rate) and $500–$1000 ($124M Lost GMV, 57.53% drop rate). Above $1000 items add another $84M in losses.

Temporal Concentration: Leakage Timing by Lost GMV (Heatmap Diagnostic)

The Day vs Hour Matrix heatmap was specifically modeled on Total Lost GMV ($) across all 24 hours (0–23) to pinpoint exact financial leakage windows rather than mere browsing volume. A careful inspection reveals two

dominant temporal dynamics:
1. Sustained Weekend Surge (Friday Hours 2–13 & Saturday Hours 4–8): The broad band of heavy revenue leakage occurs consistently on Friday morning through afternoon (especially Hours 02:00 to 13:00) and spills over into Saturday morning (Hours 04:00 to 09:00).
2. Thursday Hour 18:00: The heatmap exhibits a singular extreme dark navy outlier at Thursday 18:00 (6:00 PM).

3. Strategic Recommendations: What the Product Team Should Fix First

To maximize immediate GMV recovery, product and engineering roadmaps should prioritize high-ticket checkout friction, automated temporal retargeting, and SKU-level friction reduction:

Priority 1: High-Ticket Checkout Friction Reduction (Target: $150–$1,000+ Items)

Deploy Buy Now, Pay Later (BNPL) and flexible 0% interest financing (No-Cost EMI) directly on cart and payment summary screens for items over $150. Because Apple and Samsung devices driving $224M+ in losses cause severe final-step payment hesitation, offering transparent installment schedules significantly lowers psychological purchase resistance.

Priority 2: 1-Click Express Checkout & Digital Wallets (Target: Apple & Top SKUs)

Enable one-click express payment gateways (Google Pay, UPI, and saved biometric cards) on top leaking SKUs such as Apple SKU 1005115 ($24.48M loss) and SKU 1005105 ($12.43M loss). Eliminating multi-step billing and shipping address forms removes friction right at the tipping point of conversion.

Priority 3: Promotional Surge & Timed Retargeting Engine (Target: Thursday 18:00 & Weekend Windows)

Capitalize on the specific temporal patterns revealed by the heatmap:
• For major rollouts (such as Thursday 18:00 spikes), ensure payment gateway elasticity, instant card discount application, and server concurrency to avoid checkout freeze.
• Automated Retargeting: Trigger automated recovery push notifications and emails within 30 to 60 minutes for carts abandoned during the Friday–Saturday surge.

Priority 4: Checkout Trust Badges & Transparent Pricing (Target: Premium Tech)

Display clear return guarantees, official manufacturer warranty seals, and all-inclusive pricing (taxes and delivery fees upfront) inside the cart overlay. Eliminating unexpected last-minute fee shocks directly addresses the 57%+ drop rate on mid-to-premium products.

4. Technical Implementation & Data Pipeline Architecture

The project was executed through an end-to-end data pipeline to ensure data integrity, speed, and analytical rigor:

1. **Python Preprocessing:** Processed 67.5M raw records in Pandas, dropped 10 null session rows, eliminated 100,519 duplicate click events, parsed category strings into 3 distinct hierarchies (main, sub, product type), and saved to high-compression Parquet.

2. **MySQL Data Warehouse:** Streamed 67.4M clean records into `ecommerce_events` using chunked batch insertion (100k). Engineered indexed aggregation tables (`agg_funnel_summary`,

`agg_platform_lost_gmv_summary`, `agg_category_brand_funnel`,`agg_price_tier_funnel`, `agg_temporal_abandonment_patterns`, and `agg_top_abandoned_products`).

3. Power BI Executive Dashboard: Constructed an executive dashboard answering the business question.

5. Executive Metric Scorecard

Dashboard KPI Metric	Calculated Value	Business Interpretation & Significance
Total Cart Sessions	1,743,342 (1.74M)	Total volume of user sessions that demonstrated purchase intent
Completed Purchases	648,364 (648.36K)	Sessions successfully progressing to final payment confirmation
Cart Abandonment Rate	62.81%	Proportion of prospective buyers failing to complete checkout
Potential Lost GMV	$381,902,347 ($381.90M)	Revenue trapped in abandoned carts (item price x drop sessions)
Primary Leaking Category	Electronics ($258.37M)	67.6% of all lost platform revenue concentrated in single department
Top Leaking Brand	Apple ($160.55M)	Over 42% of total platform-wide lost GMV attributed to Apple products
Core Price Exposure Tier	$150 – $500 ($132.00M)	Core consumer sweet spot with largest cumulative dollar leakage
Matrix Heatmap Focus Metric	Total Lost GMV ($)	Color saturation and cell density driven by dollar impact across 24 hours (0–23)
Peak Abandonment Windows	Friday (02:00–13:00) & Saturday (04:00–09:00) | Thursday 18:00	Timing leakage by lost GMV